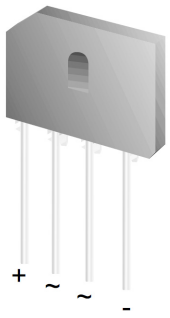
Bridge Rectifiers

GBU4A - GBU4M



SIP4
CASE 127EL

Features

- Glass−Passivated Junction
- Surge Overload Rating: 150 A Peak
- Reliable Low−Cost Construction Utilizing Molded Plastic Technique
- Ideal for Printed Circuit Board
- UL Certified: UL #E258596

PACKAGE MARKING AND ORDERING INFORMATION

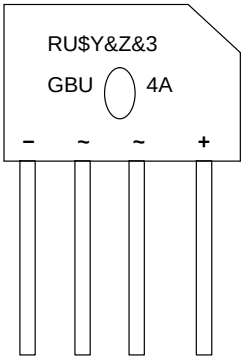
Part Number	Marking	Package	Packing Method
GBU4A	GBU4A	GBU 4L	Rail
GBU4B	GBU4B		
GBU4G	GBU4G		
GBU4J	GBU4J		
GBU4K	GBU4K		
GBU4M	GBU4M		

DISCONTINUED (Note 1)

GBU4D	GBU4D	GBU 4L	Rail

1. **DISCONTINUED:** This device is not recommended for new design. Please contact your **onsemi** representative for information. The most current information on this device may be available on www.onsemi.com.

MARKING DIAGRAM



RU	= UL Marking
$Y	= **onsemi** Logo
&Z	= Assembly Plant Code
&3	= Numeric Date Code
GBU4A	= Specific Device Code

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ABSOLUTE MAXIMUM RATINGS (T_A = 25°C unless otherwise noted) (Note 2)

Symbol	Parameter		Value							Units
			4A	4B	4D	4G	4J	4K	4M	
V_{RRM}	Maximum Repetitive Reverse Voltage		50	100	200	400	600	800	1000	V
V_{RMS}	Maximum RMS Bridge Input Voltage		35	70	140	280	420	560	700	V
V_R	DC Reverse Voltage (Rated V_R)		50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current	T_A = 100°C	4.0							A
		T_A = 40°C	3.0							A
I_{FSM}	Non−Repetitive Peak Forward Surge Current 8.3 ms Single Half−Sine−Wave		150							A
T_{STG}	Storage Temperature Range		−55 to +150							°C
T_J	Operating Junction Temperature		−55 to +150							°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

2. These ratings are limiting values above which the serviceability of any semiconductor device may be impaired.

THERMAL CHARECTERISTICS (T_A = 25°C unless otherwise noted)

Symbol	Parameter	Value	Units
P_D	Power Dissipation	8	W
$R_{\theta JA}$	Thermal Resistance per Leg, Junction to Ambient (Note 3)	19	°C/W

3. Device mounted on PCB with 0.5 × 0.5 inch (12 × 12 mm)

ELECTRICAL CHARACTERISTICS (T_J = 25°C unless otherwise noted)

Symbol	Parameter		Value	Units
V_F	Forward Voltage, per Element at 4.0 A		1.0	V
I_R	Reverse Current, per Element at Rated V_R	T_A = 25°C	5.0	μA
		T_A = 125°C	500	μA
I^2t	I^2t Rating for Fusing		93	A^2s

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

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TYPICAL PERFORMANCE CHARACTERISTICS

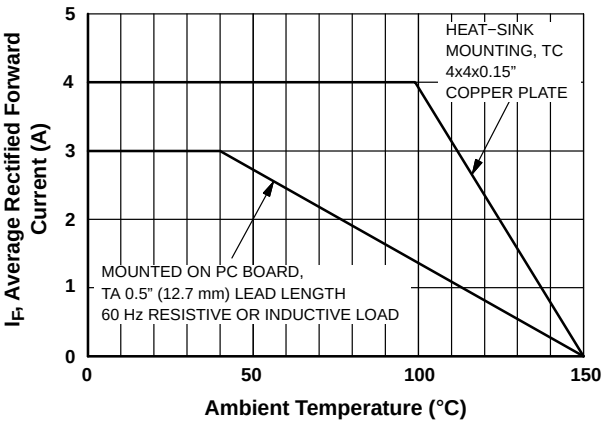


Figure 1. Forward Current Derating Curve

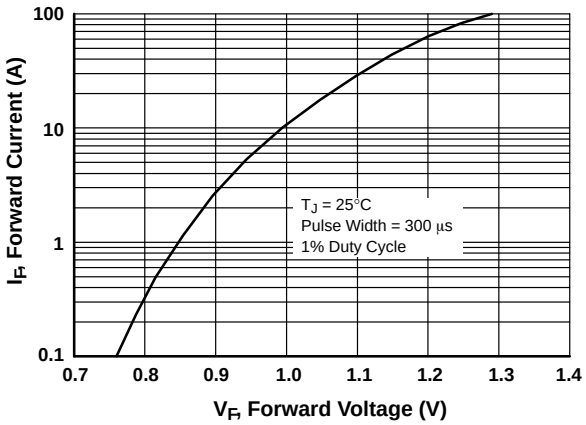


Figure 2. Forward Voltage Characteristics

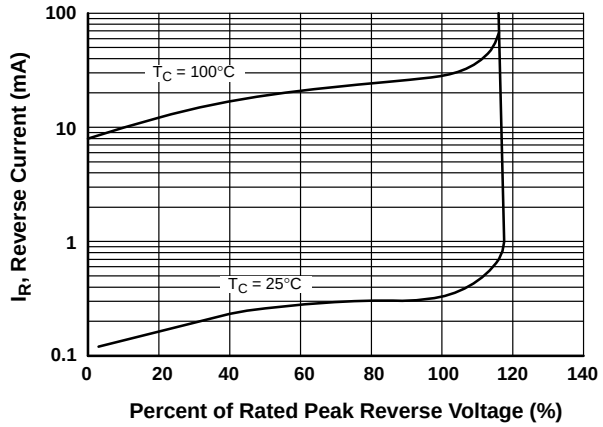


Figure 3. Reverse Current vs. Reverse Voltage

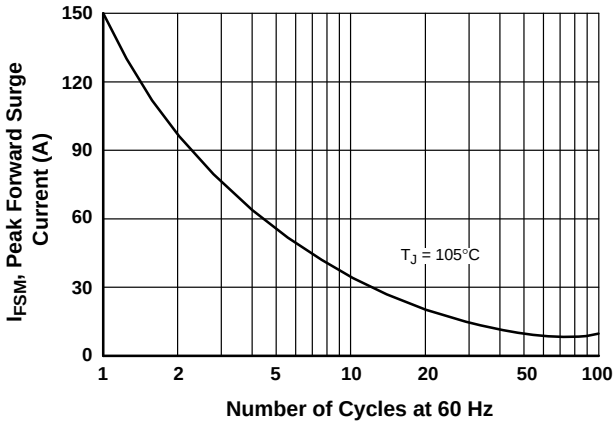


Figure 4. Non−Repetitive Surge Current

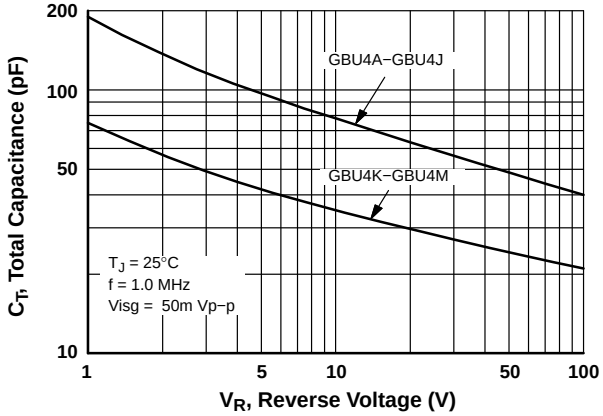
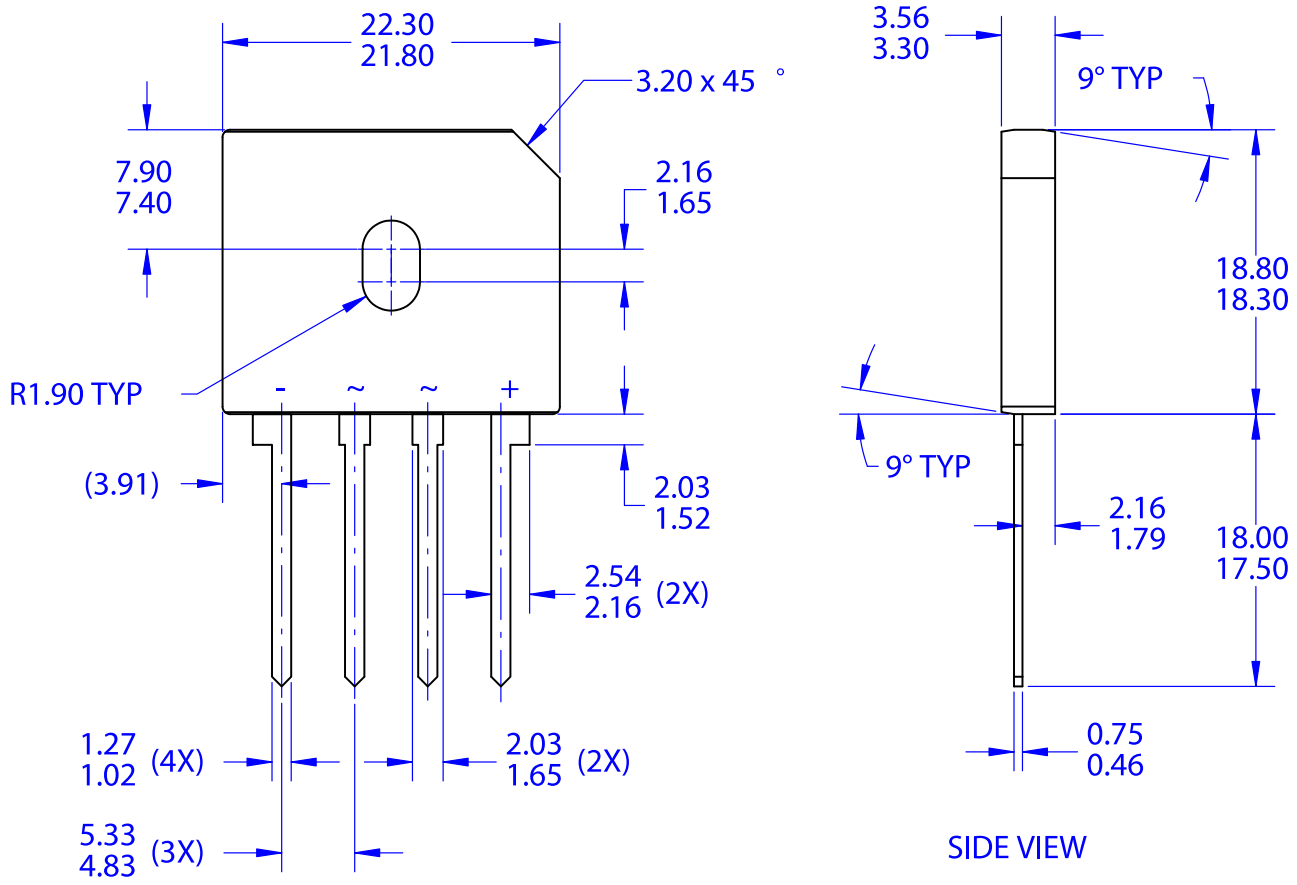


Figure 5. Total Capacitance

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